3 **(a)** e.g. two objects of different masses at same temperature (M1)
same material would have different amount of heat (A1)
e.g. temperature shows direction of heat transfer (M1)
from high to low regardless of objects (A1)
e.g. when substance melts/boils (M1)
heat input but no temperature change (A1)
any two, M1 + A1 each, max 4 .... [4]

(b) **(i)** energy losses (to the surroundings) M1
either increase as the temperature rises
or rise is zero when heat loss = heat input .... A1 [2]

(ii) idea of input <u>power</u> = maximum <u>rate</u> of heat loss C1
power = $m \times c \times \Delta\theta / \Delta t$
$54 = 0.96 \times c \times 3.7 / 60$ C1
$c = 910\ \text{J kg}^{-1}\ \text{K}^{-1}$ .... A1 [3]

[Total: 9]

4 **(a)** **(i)** amplitude = 0.2 mm A1 [1]